Morphology (2025) 35:545–568
https://doi.org/10.1007/s11525-025-09446-1

Diversity of schemas in English bahuvrihi compounds

Petr Kos[1]

Received: 31 March 2025 / Accepted: 19 June 2025 / Published online: 30 June 2025

Abstract
The present study examines a dataset of English bahuvrihi compounds with the right-hand constituent *head*, drawn from the *Araneum Anglicum Maius* corpus and other supplementary sources, with the aim of investigating how this specific section of the lexicon is internally structured. Using the theoretical framework of Relational Morphology developed by Jackendoff and Audring, the study seeks to identify the diversity of schemas that can be abstracted from the dataset. It explores the idiosyncratic features of individual schemas and demonstrates their implications for structuring the lexicon (relational function) and the formation of new lexemes (generative function). Regarding the relational role, the schemas exhibit a hierarchical structure, ranging from low-level highly specific schemas to the most abstract ones at the highest level. The schemas also function as categories comprising prototypical and peripheral members and exhibit fuzzy boundaries. These fuzzy boundaries are manifested as overlaps between two schemas, where some lexemes may belong to both simultaneously. Additionally, it is argued that certain series of lexemes display specific patterns in their conceptualization, which may further contribute to structuring the lexicon and to the formation of novel lexemes.

Keywords Bahuvrihi compounds · Schemas · Relational function · Generative function

1 Introduction

Within the framework of Relational Morphology, the lexicon comprises not only fully specified lexemes but also schemas abstracted from them. In Jackendoff and Audring (2020a), the theoretical aspects of schemas and their role in the structure of the lexicon are discussed and illustrated through representative examples based on a limited number of lexemes.

✉ P. Kos
kos@ff.jcu.cz

1 University of South Bohemia in České Budějovice, Branišovská 31a, 37005 České Budějovice, Czech Republic

The present study seeks to investigate the nature of schemas on a substantially larger dataset that would be both comprehensive and unified. The dataset is drawn from a corpus and supplementary sources, with the aim of identifying the schemas that can be abstracted from the data and exploring how these schemas are structured within the lexicon. The selected form under investigation is that of English bahuvrihi compounds with the right-hand constituent *head*. Although these compounds superficially share the same form and internal structure, they exhibit considerable heterogeneity, revealing a variety of specific schemas within this particular "island" of the lexicon.

This case study investigates the idiosyncratic features of the individual schemas that can be abstracted from the dataset and demonstrates how these schemas function both in structuring the lexicon (the relational role) and in the formation of new lexemes (the generative role). As such, the study serves as a probe into a specific section of the lexicon, with the potential to generalize the implications on the nature of schemas beyond the analysed form.

The article is structured as follows: Sect. 2 outlines the theoretical framework applied in the study; Sect. 3 describes the data collection process; Sect. 4 addresses the processing of the acquired data; Sect. 5 presents the analysis of individual schemas abstracted from specific series in the dataset; Sect. 6 examines the implications for the relational function of the schemas; and Sect. 7 discusses their generative function.

2 Theoretical framework

The approach to word structure and word-formation patterns applied in the present study is based on the Relational Morphology theory developed by Ray Jackendoff and Jenny Audring (2020a), with some insights from the related Construction Morphology theory developed by Geert Booij (2010). In their approach, the lexicon of a speaker comprises not only fully specified words but also schemas abstracted from them. A schema thus expresses parallelism among a series of words that are the same in some respect but differ in others. Schemas are stored in the same format as words, with the exception that schemas contain variables. All schemas have a relational role, providing structure to the lexicon, and only some schemas also have a generative role, providing patterns for the formation of new lexemes. A schema has a declarative character, worded as "*this* is a possible structure" (Jackendoff & Audring, 2020a: 28, italics in original).

The format of a lexical item, both a fully specified word and a schema, follows that of the Parallel Architecture framework (Jackendoff, 2002). In this format, three levels of a lexical item, namely the semantic, morphosyntactic, and phonological, are linked. The association between the three levels in the long-term memory is provided by interface links, marked by an index, as in the notation of the example *bighead* (fish; OED) (1), taken from the analysed dataset.

(1) Semantics: [FISH WHOSE $(\text{HEAD})_2$ HAS FEATURE $(\text{BIG})_1$; *HYPOPHTHALMICHTHYS NOBILIS*$]_3$
Morphosyntax: $[_{\text{N}}\text{A}_1\ \text{N}_2]_3$
Phonology: /ˈbɪg$_1$ hed$_2$/$_3$

On the semantic level, two types of meaning are given: the structural meaning, which refers to the semantic structure of the complex word, and the lexical meaning, which refers to the concept denoted by the word (here notated by the Latin term for the particular species of fish). The morphosyntactic level states the word-classes of individual components, and the phonological level gives their pronunciation including the position of the primary stress in the whole compound. The indices 1, 2, and 3 signal the interface links between the individual components of the compound, and the index N on the morphosyntactic level states the word-class of the whole complex lexeme. The category of the concept denoted by the compound indicated on the semantic level, FISH, has no index and is thus not linked to any constituent of the compound, which determines the compound's exocentric character.

From other lexemes sharing the same pattern, such as *snakehead* (plant), *steelhead* (fish), *thickhead* (bird; all OED), etc., as well as *longleaf* (tree), *longfin* (fish), *spoonbill* (bird; all OED), etc., we can abstract schema (2).

(2) Semantics: [NATURAL ORGANISM WHOSE (BODY PART)$_X$ HAS FEATURE (Y)$_Y$]$_Z$
Morphosyntax: $[_N N/A_Y\ N_X]_Z$
Phonology: /ˈ...$_Y$ ...$_X$/$_Z$

Unlike in the notation for a fully specified lexeme (1), the semantic level of schema (2) specifies the structural meaning only, with the two variables given in the round brackets. The variable with index Y is semantically underspecified as it may be instantiated with any relevant feature, and the variable with index X is semantically restricted to BODY PART. Since all the lexemes that instantiate schema (2) appear to share the same semantic and morphological pattern, including their connotational meaning being stylistically neutral, there is no reason to assume that different types of organisms would have their own specific schemas; the category of the possible referents thus covers all natural organisms. Consequently, the category in the notation of fully specified lexemes can be a hyponym of the category in the schema. On the morphosyntactic level, the left-hand constituents of fully specified lexemes can be either nouns or adjectives, which is notated in the schema with a slash. On the phonological level, the three dots represent arbitrary phonological strings, and the symbol "ˈ" signals the position of the primary stress.

The notation of the lexical items (both fully specified lexemes and schemas) in this study thus corresponds to the one in Jackendoff and Audring (2020a). There is, however, a specific pattern in the notation of the semantic level, which deserves further clarification.

As suggested above, the notation distinguishes between the structural and lexical meanings. The structural meaning is motivated by the semantics of the components of the lexical item only, and the lexical meaning captures the rich semantics of the concept the lexeme denotes. Morphologically motivated words have both the structural and lexical meanings, schemas have the structural meaning only, and unmotivated words have the lexical meaning only.

The notation of the structural meaning is in the form of a subordinate restrictive relative clause. The syntactic head of the clause refers to the category the concept is assigned to. The category may vary in specificity, the most general being that of the basic cognitive category, i.e., for the word class of nouns, it is the category THING (see Langacker, 1987, 1991). In (1) and (2), the category has no index as it has no representation on the morphosyntactic and phonological levels. However, in endocentric compounds, it is represented by their right-hand constituents as shown in the notation for *butterfish* (OED) (3).

(3) Semantics: [(FISH)$_5$ WHICH IS ASSOCIATED WITH (BUTTER)$_4$; *PSENOPSIS ANOMALA*]$_6$
Morphosyntax: [$_N$N$_4$ N$_5$]$_6$
Phonology: /ˈbʌtə$_4$ fɪʃ$_5$/$_6$

(3) instantiates schema (4) for endocentric NN compounds.

(4) Semantics: [(Y)$_Y$ WHICH IS ASSOCIATED WITH (X)$_X$]$_Z$
Morphosyntax: [$_N$N$_X$ N$_Y$]$_Z$
Phonology: /ˈ. . .$_X$ . . .$_Y$/$_Z$

Within the relational role of schemas, the category, even if it has no representation on the morphosyntactic and phonological levels as in (2), contributes to structuring the lexicon through relational links that group together lexemes sharing the same category, and within their generative role, it is instrumental in matching the initial conceptualization of the named concept with a suitable schema, as further described in Sect. 7.

The restrictive relative clause implies that the further specification is in a subordinate position to the category. It expresses a feature that distinguishes the concept denoted by the lexeme from other members of the category the concept belongs to. So, going back to the example *bighead* (1), the feature that *Hypophthalmichthys nobilis* has a big head distinguishes this species from other members of the category, i.e., from other species of fish.

The two positions in the schemas for the type of compounds under analysis in this study can have a varying level of specificity. They can be:

a) semantically open variables – there is no restriction on the semantic level, apart from the highest conceptual categories corresponding to nouns or adjectives (notated on the morphosyntactic level). As for the notation on the semantic level, such variables are signalled by round brackets, and the semantic underspecification is notated by the same letter as the constituent's index, e.g. (Y)$_Y$.
b) semantically restricted variables – the position can be instantiated with a limited set of lexemes only. The semantic specification is given in round brackets, e.g. (BODY PART)$_X$ or (ITEM OF INTEREST)$_Y$.
c) the position is fixed and literal – the position is fully specified but has not lost its semantic value. For instance, all terms for Harley Davidson engines, such as *knucklehead* and *panhead* (both AAM/Wikipedia), comprising a unified series have *head* in the right-hand position (which makes it fixed in the abstracted schema), but in all cases the right-hand constituent refers to the upper part, the head, of the engine. In other words, it is fixed but not idiomatic. It is notated by an index on the semantic level but not in round brackets, e.g., HEAD$_X$.

d) the position is fixed and idiomatic – these are instances when all the members of the series share the same constituent, but the constituent has no direct literal representation on the semantic level. Booij (2010: 57) calls such instances as "constructional idioms". For example, all terms for enthusiasts sharing the same structure, such as *gearhead* and *sneakerhead* (both AAM/OED), have *head* in the right-hand position, which, however, has become semantically impoverished. Such "affixoids" (see Booij, 2010: 57) have representation on both the morphosyntactic and phonological levels (linked by an index) but not on the semantic level. In this respect they resemble affixes, which do not have an index on the semantic level either (see Jackendoff & Audring, 2020a: 130, for explanation), but cannot be seen as affixes since the fixed position may form more abstract, higher-level, series with lexemes in which the position is a variable.

Schemas are hierarchically structured, meaning that the lexicon comprises schemas on different levels of relative specificity, broader or narrower, in respect to one another. Schemas of different specificity can be constructed in a "bottom-up" or a "top-down" manner. The "bottom-up" perspective suggests that "relatively narrow schemas give rise to broader ones" (Jackendoff & Audring, 2020a: 227), and "top-down" approach implies the construction of a narrower schema inside a broader one.

The lexicon comprises both fully specified lexemes and abstract schemas. In other words, the fully specified lexemes do not necessarily disappear from the long-term memory once the schema is abstracted, and they are "encoded in their entirety, even where redundant" (Jackendoff & Audring, 2016: 476). One of the consequences is that the idiosyncratic semantic specifications encoded in fully specified lexemes that are not reflected in the abstracted schema are still present in the memory and can play a role within both the relational (see Sect. 6) and generative (see Sect. 7) functions of schemas.

3 Data collection

As stated above, the aim of the present study is to investigate the nature of schemas on a large, unified dataset. To this end, a type of lexemes demonstrating both semantic diversity and unity of form was selected for analysis, namely English bahuvrihi compounds. The unifying feature of these compounds is that the right-hand constituent denotes a part of the entity denoted by the whole compound. Due to a high variability of parts that can form a whole and in order to acquire as complete a list of lexemes of one type as possible, the search was limited to a specific part in the right-hand position of the compounds – *head*. The suitability of *head* lies in the fact that it denotes a highly salient part of humans and animals and at the same time it is used metaphorically for top parts of inanimate objects. This ensures a semantic variety within a unified sample of data.

For the sake of completeness, the aim of the data collection was to acquire as many lexemes of this form as possible. The primary tool for the acquisition of forms was the Araneum Anglicum Maius corpus (AAM; 1,2 G tokens; see Benko, 2014, for a detailed description) with a subsequent search in other internet sources and dictionaries, such as Google, Wikipedia, OED, Urban Dictionary (UD), and Wiktionary. The

Table 1 Contribution of different sources to the dataset (forms only)

	Type frequency	%
AAM	265	80
UD	20	6
Google	16	5
Wikipedia	14	4
OED	13	4
Wiktionary	2	1
Total	330	100

search in the supplementary sources was instrumental in identifying lexemes that are too specific and/or peripheral to find their way into the corpus, such as those belonging to slang and technical jargon. It focused on other possible forms (e.g., possible items of interest in the left-hand position in the category of enthusiasts) or on other possible referents within identified categories (e.g., a Google search for "types of fasteners", beyond those identified in the corpus). Overall, the search provided a total of 330 forms (type frequency). The contribution of different sources to the dataset is given in Table 1, with 80% of the number coming from the AAM corpus and 20% from the other sources.

In order to compile as comprehensive a dataset as possible, not restricted to the meanings attested in the corpus, the forms were paired with all recorded meanings using various available dictionaries, such as, but not restricted to, OED, Collins, Wiktionary, and UD (for peripheral non-standard meanings). The overall acquired number of bahuvrihi compounds with the right-hand constituent *head* (pairs of form and meaning, type frequency) is 540. The difference in frequency between the sole forms and those paired with their meanings is given by the fact that one form may have a number of meanings; for example, *whitehead* (AAM) may denote an old person (UD), a blond woman, a cocaine addict (both Green's Dictionary of Slang), various species of birds, and a type of acne (both OED). Each pair of form and meaning is thus seen as a separate item, a separate lexeme.

The lexemes manually eliminated from the dataset were those that do not have an exocentric character, i.e., endocentric compounds, such as *arrowhead* and *warhead*, and those that do not otherwise demonstrate the internal structure of the type of compounds under analysis, i.e., overall metaphors, such as *doghead* (the hammer of a gunlock having the shape that resembles a dog's head), eponyms, such as *whitehead* (a torpedo invented by Robert Whitehead), and toponyms, such as *Sandhead* and *Spithead* (British place names).

4 Data processing

The processing of the acquired dataset of 540 bahuvrihi compounds with the right-hand constituent *head* imitates the mental process in the mind of the speaker. Since in a speaker's mind "the acquisition of a schema has to be item-based" (Jackendoff & Audring, 2016: 478), schemas are abstracted from series of existing lexemes that share some common properties on the basis of the same-except principle and, as such, express parallelism among all the lexemes in the series. The dataset is thus conceived as part of the mental lexicon of an idealized "superhero" speaker which comprises all lexemes of the language including all regional and stylistic varieties (British and American English, other regional varieties, technical jargon, biological classification, slang, etc.). Mental lexicons of individual speakers are, of course, much more fragmented as no speaker of the language possesses such capacity.

Since all lexemes in the dataset have the same morphological structure, in that the compound as a whole is a noun, the right-hand constituent of the compound is the noun *head*, and the left-hand constituent is typically a noun or an adjective, the search for shared commonalities focused on the semantic level in the following steps.

First, the dataset was divided into major categories the referent belongs to, i.e., whether the referent is a human, a non-human natural organism, or an inanimate entity. Second, within individual categories the search was aimed at identifying any dominant subseries whose abstraction would justify the existence of a more specific subschema (see Booij, 2010: 55). Third, as for the right-hand constituent, it was determined whether it is a variable or a fixed constituent. In order to do that, additional search within each (sub)series was carried out to find out whether other lexemes that belong to the same (sub)series but have a different part of the whole in the right-hand position (e.g., *mouth* as in *bigmouth* (boastful person) or *back* as in *silverback* (gorilla; both OED)) can be identified. The search focused either on expected parts of the whole for a given category (e.g., body parts for humans) in the corpus and established dictionaries (see above) or on other members of specific subseries (e.g., Harley-Davidson engines or fish lures) making use of Google search engine (search for "types of X"). If instances of such lexemes were identified, the right-hand constituent was determined to be a variable with the consequence that the series cannot be considered complete. Fourth, the semantic motivation of the left-hand constituent was analysed in order to find out whether there are any tendencies in the conceptualization of the concepts/referents vis-à-vis their names.

For the purposes of the present study, the major parameters for the decision whether an identified series is to be schematized were the possibility of determining a unified structural meaning on the semantic level (see the discussion on ethnic groups in Sect. 5.1) as well as the size of the series. If the series consisted of a limited number of lexemes only, further parameters were applied, namely to what extent the category of the referents is distinct from that of other members of the larger series (e.g. acne is distinct from all other inanimate referents in not being a man-made object; see Sect. 5.3.4) and to what extent the lexemes in the series appear to be entrenched in the long-term memory, which is proportional to their resting activation, i.e., their frequency of use. The latter three parameters are assumed to be linked to prominence, in that the larger the series is, the more prominent the pattern is, a clearly distinct category is prominent within a more general category, and lexemes with higher resting activation are prominent in comparison with rarely used ones. The underlying assumption about the role of prominence of lexemes in establishing a schema is that a schema, as an additional lexical item in the memory, represents an extra "cost" to memory, while also providing cognitive benefits (see Jackendoff & Audring, 2020a: 80), and only series prominent enough "are worth" the cost.

The above-mentioned steps and assumptions have led to the proposed design of a schema for each distinct series within the dataset (see Sect. 5). It needs to be stressed, however, that the resulting schemas are tentative only since “it is not necessarily the case that all language users make the same subgeneralizations. Schemas are based on lexical knowledge, and this type of knowledge varies from speaker to speaker. Hence, speakers may also differ in the number and types of schemas that they deduce from their lexical knowledge” (Booij, 2010: 89).

Also, individual dominant series may still be rather heterogeneous. As such, they are perceived as categories with prototypical and peripheral members. Their description thus focuses on the prototypical characteristics in order to capture the general tendencies, and the peripheral ones are often merely mentioned for illustrative purposes.

5 Individual series

This section provides a description of morphological and semantic properties of each distinct series in the dataset and the resulting tentative schema.

5.1 Humans

The series of bahuvrihi compounds with *head* as the right-hand constituent denoting humans comprises 376 lexemes in total. The members of the group vary in their denotations; nevertheless, within the series some dominant specific subseries may be abstracted, as shown in Table 2.

The compounds comprising the miscellaneous series exhibit a general pattern in the formation of bahuvrihi compounds denoting humans, captured in schema (5). This schema can be considered as a general, higher-level, schema for humans with respect to the more specific subschemas, (6)–(9), below.

(5) Semantics: [PERSON WHOSE (BODY PART)$_X$ IS ASSOCIATED WITH (Y)$_Y$]$_Z$
Morphosyntax: [$_N$N/A$_Y$ N$_X$]$_Z$
Phonology: /ˈ. . .$_Y$ . . .$_X$/$_Z$

Prototypically, the lexemes instantiating schema (5) denote humans. Included as peripheral members, however, are also lexemes denoting non-human referents that exhibit anthropomorphic features, such as toys/dolls (e.g., *splithead*, doll; AAM/Wiktionary) or characters in cartoons or computer games (e.g., *gogglehead*, manga character; AAM/UD). Those denoting humans are prototypically derogatory, with the exception of *chillhead* as a term for “a generally worthwhile human being” (UD) and terms denoting rugby players, *loosehead* and *tighthead* (both AAM/OED). The terms for non-human referents are stylistically neutral, which underpins their peripheral nature.

As the series also comprises lexemes that have other body parts than *head* in the right-hand position, such as *tenderfoot* (inexperienced person), *highbrow* (intellectual person), or *loudmouth* (loudly speaking person; all OED), the right-hand constituent proves to be a variable. The left-hand constituent may refer to a physical feature, as

Table 2 Frequency of distribution of specific series denoting humans with examples

	Freq.	%	Examples
Humans – miscellaneous	139	37	*bighead* (AAM/OED),[1] *bluehead* (AAM/UD)
"Stupid" people	100	26	*dumbhead, muttonhead* (both AAM/OED)
Drug abusers	44	12	*acidhead* (AAM/OED), *brownhead* (UD)
Enthusiasts	93	25	*jazzhead* (AAM/Wiktionary), *petrolhead* (AAM/OED)
Total	376	100	

[1]The bracket states the source of the form on the left and the source of the meaning on the right. If both the form and meaning come from the same source, only the one source is stated.

in *slaphead* (bald person; OED) or *redhead* (person with red hair), to a person's character, as in *bighead* (conceited person) or *pighead* (stubborn person; all AAM/OED), or to a behavioural trait, such as *zebrahead* (a white man who exclusively dates people with dark skin; UD). These three types of reference also have an influence on the perception of the head in the compounds. Those referring to a person's physical feature see the head as a physical entity (either literally or metonymically referring to the hair or headwear, as in *conehead* – a bishop in the Catholic Church; UD), those referring to a person's character metonymically as a locus of mental capacity, and those referring to a behavioural trait metonymically as the whole person. Due to its variability, the semantic relation between the two constituents of the compounds is underspecified, notated as "is associated with".

The series also includes a more specific subseries referring to humans as members of ethnic groups. What makes them specific is that the character of individuals in the denotation of these lexemes is backgrounded, and foregrounded is some stereotypical aspect associated with the whole group. Thus, *dothead* (OED) refers to a person of South Asian, particularly Indian, descent even though individual members of the ethnic group may not wear the bindi, the traditional forehead marking. The same applies to *cheesehead* (AAM/OED) as a term for inhabitants of Wisconsin, where cheese is rather a stereotypical reference to the whole state than to its individual inhabitants.

This subseries comprises two types of lexemes. The first one refers to the head as a body part, either to the head itself as in *flathead* (a member of some indigenous peoples of North America, the term motivated by shape; AAM/OED) or *brownhead* (a person of Southeastern Asian origin, the term motivated by complexion colour; UD) or metonymically to its hair, *woollyhead* (a person of African or Arabic origin), or a headwear, as in *raghead* (a person of Asian origin; both AAM/OED). The second type comprises lexemes which refer to some prototypical feature associated with the ethnic group, such as *pizzahead* (an Italian), *fishhead* (a person of Southeastern Asian origin; both AAM/Wiktionary), or *canucklehead* (a Canadian, *canuckle* being a term for a Canadian word game, see Google; AAM/UD).

However, this specific subseries of terms denoting members of ethnic groups does not apparently lead to the emergence of a more specific subschema, since the two types of lexemes comprising the series do not have a unified internal structure. Instead, we may assume a specific pattern of conceptualization within schema (5) only, where the whole group is mentally accessed through a prototypical member of the group, namely the MEMBER OF CATEGORY FOR CATEGORY metonymy.

The dataset also includes specific microseries, such as *breezehead*, *windhead*, and *galehead* (cf. the definition of *windhead* (AAM/UD) as "a person who does not think as fast as a galehead but does think faster than a breezehead") or the above-mentioned terms for types of rugby players, *loosehead* and *tighthead*. Such microseries open the question whether they lead to the emergence of highly specific subschemas. Given their small size, their peripheral character, and the fact that they fully comply with the general schema (5), i.e., they have a low level of prominence, we may assume that storing an extra lexical item (i.e., a schema) abstracted from such short series would lead to an unnecessary burden to the long-term memory of the speaker, so such "small islands" are seen as analogical formations not requiring a specific schema in the structured lexicon.

The subsections below discuss individual dominant subseries denoting humans which instantiate specific subschemas.

5.1.1 "Stupid" people

All lexemes comprising subschema (6) for "stupid" people are deliberate terms of abuse and highly derogatory. As Bauer and Huddleston (2002: 1652) suggest, they may also be used as vocatives, as in "*Hey, sillyhead!*". The series comprises 100 lexemes in the dataset, such as *blockhead*, *dumbhead*, and *woodenhead* (all AAM/OED).

(6) Semantics: [PERSON WHOSE INTELLECTUAL CAPACITY$_X$ IS ASSOCIATED WITH (Y)$_Y$]$_Z$
Morphosyntax: [$_N$N/A$_Y$ N$_X$]$_Z$
Phonology: /ˈ...$_Y$ hed$_X$/$_Z$

Even though there are other terms of abuse in the form of bahuvrihi compounds with other right-hand constituents than *head*, as those with *brain*, as in *birdbrain* (OED), *dirtbrain* (Wiktionary), and *pissbrain* (Wiktionary), and we can thus also assume the existence of a more general subschema for "stupid" people, the subseries with *head* is so substantial that we can surmise the right-hand constituent to be fixed as *head* for these lexemes. The link (notated by index X) between the semantic level and the other two levels is still apparent, since all terms in this series refer metonymically to the referent's intellectual capacity, whose location is in the head.

The reference to "stupidity", expressed by the left-hand constituent, may be direct through adjectives, as in *stupidhead* (AAM/OED) or *sillyhead* (AAM/Wiktionary), nouns, as in *doodlehead* (*doodle* denoting a fool, see OED; AAM/Wiktionary), or through downright vulgarisms, as in *fuckhead* or *shithead* (both AAM/OED). Another rather straightforward reference to "stupidity" is the comparison with "dull" animals, as in *muttonhead* or *oxhead* (both AAM/OED), or "dull" commonplace objects, as in *bananahead* or *shovelhead* (both AAM/UD).

More interestingly, the lexemes also demonstrate certain tendencies in the conceptualization of "stupidity" via conceptual metaphors (see Barcelona, 2008, for a more comprehensive treatment). Lack of intelligence may thus be conceptualized as STUPIDITY IS SLOWNESS, i.e. demonstrating slow thinking, manifested in the left-hand constituent as a slow-flowing substance, as in *saphead* and *thickhead* (both AAM/OED) or in association with slow-moving obese people, as in *fathead* (AAM/OED), as STUPIDITY IS HARDNESS, i.e. intellectually impenetrable, as in *hardhead, bonehead*, or *brickhead* (all AMM/OED), as STUPIDITY IS MUSHINESS, i.e. lacking intellectual quality, as in *puddinghead, doughhead*, or *cheesehead* (all AAM/OED), and as STUPIDITY IS EMPTINESS, i.e. lacking brain, as in *airhead, bubblehead* (both AAM/OED), or *hollowhead* (AAM/UD). Lack of intelligence is also associated with a small brain, as in *gooberhead* (AAM/UD) and *pinhead* (AAM/OED).

Included in the series, as peripheral members, are also two terms for non-human referents, namely *jarhead* and *jughead* (terms used in the U.S. Army as slang derogatory reference to mules; both AAM/OED). The justification for the inclusion of these two lexemes referring to an animal in the series denoting humans lies in their derogatory character, which is otherwise absent in terms for natural organisms, in the fact that they are slang words rather than classificatory terms for natural organisms (see Sect. 5.2 below), and in the fact that they follow a pattern of conceptualization applied in some other terms in the series, namely the conceptual metaphor STUPIDITY IS EMPTINESS (cf. the left-hand constituents *jar* and *jug* denoting empty containers).

5.1.2 Drug abusers

The series of lexemes instantiating subschema (7) that denote people abusing addictive substances, such as alcohol and illegal drugs, comprises 44 lexemes in the dataset, such as *hophead* (opium user), *pothead* (marijuana user), and *rockhead* (cocaine user; all AAM/OED).

(7) Semantics: $[\text{PERSON WHO ABUSES (DRUG)}_{Y}]_{Z}$
Morphosyntax: $[_{N}N_{Y}\ N_{X}]_{Z}$
Phonology: $/^{\prime}\ldots_{Y}\ \text{hed}_{X}/_{Z}$

These lexemes are all derogatory in their reference. In schema (7), the right-hand constituent is fixed as *head* and has no direct link to the semantic level, which gives it the character of an affixoid (see Sect. 2). As for the relatedness of this affixoid with the corresponding lexeme *head* (cf. Booij, 2010: 57), diachronically, the first coinages (year 1856 onwards, see OED) were apparently formed through the association of the effect of drug consumption on the head (brain) of the user employing the general schema for humans (5), so, synchronically, a loose link to the lexeme *head* may still be assumed, which also justifies the relatedness of schema (7) to the general one for humans (5).

Prototypically, the left-hand constituent refers to the substance abused. The reference may be direct, as in *acidhead* (LSD; AAM/OED) or *whiskyhead* (OED), or expressed metonymically through its colour, as in *brownhead* (heroin; UD) and *whitehead* (cocaine; AAM/Green's Dictionary of Slang), its origin, as in *poppyhead* (opium; AAM/UD), or its form, as in *pillhead* (AAM/OED) or *beanhead* (AAM/UD). Since the vast majority of lexemes in the series demonstrate reference to the substance used, the left-hand constituent on the semantic level of subschema (7), indexed with Y, is notated as DRUG. Moreover, there are but three instances that do not follow this pattern, namely *tinhead* (motivated by the manner of use, the drug being wrapped in foil when smoked; AAM/UD), *stonehead* (AAM/Wiktionary), and *granolahead* (motivated by the effect of the drug, a confused mind; AAM/UD). However, given their peripheral nature, they are not considered to affect the major tendency in the series of instantiating the left-hand constituent with a lexeme referring to an abused substance.

5.1.3 Enthusiasts

Closely related, in some cases even overlapping, to the lexemes denoting drug abusers are those denoting enthusiasts for some item of interest. They denote humans as devotees. The fuzzy division line between addiction and devotion can be seen, for instance, in *coffeehead* (AAM/Wiktionary x Reverso Dictionary) or *techhead* (AAM/Collins x OED) where the referents may be seen either as addicted to or as enthusiastic about coffee and technological devices, such as computers, respectively. In these lexemes, addiction and enthusiasm can be perceived as two facets of one meaning; in other words, as two elaborations of a more schematized meaning (see Langacker, 2013: 17), which could be worded as "someone excessively consuming coffee" and "someone excessively spending time on computers", respectively. Which of the two facets, the negative or the positive connotation, is foregrounded depends on the context of use, when it is given its full contextual value (see Langacker, 1999: 63). A case where this distinction has apparently led to two independent lexemes is *winehead* (AAM/UD), which, together with its synonym *wino* (Wiktionary), may refer either to a homeless person habitually drunk on wine or to a wine enthusiast or connoisseur. The justification for the independent status of the series denoting enthusiasts from that for drug abusers is mainly in its non-derogatory, or even positive, connotation. Another aspect is that the left-hand constituent need not (and in most cases does not) refer to an ingestible substance.

The series of lexemes denoting enthusiasts also comprise a dominant subgroup, those of music enthusiasts, such as *funkhead*, *pophead*, and *raphead* (all UD). Given their specific nature, it is assumed that these form a distinct series instantiating a specific subschema (see below).

Overall, the series of lexemes denoting enthusiasts comprises 93 items in the dataset, 19 of which denote music enthusiasts. The general subschema for enthusiasts is (8).

(8) Semantics: [PERSON WHO IS ENTHUSIASTIC ABOUT (ITEM OF INTEREST)$_Y$]$_Z$
Morphosyntax: [$_N$N$_Y$ N$_X$]$_Z$
Phonology: /ˈ. . .$_Y$ hed$_X$/$_Z$

Apart from referring to humans, the lexemes in the series also serve as brand names for local businesses, such as *Teahead* (a tearoom) or *Breadhead* (a bakery), websites, such as *Surveyhead* (a website offering surveys; all AAM/Google) or *Wowhead* (a website for World of Warcraft computer game fans; AAM/Google), and products, such as *Applehead* and *Cherryhead* (types of sweets; both Google). Their inclusion in the series, otherwise denoting human referents, is substantiated by the interpretation of these brand names either as "we are X-enthusiasts" or as "business/website/product meant for X-enthusiasts". So, there is a metonymic shift from enthusiasts to entities run by or produced for enthusiasts. Naming business entities with subschema (8) may also be encouraged by the positive connotation of these terms.

The right-hand constituent is fixed as *head* with no link to the semantic level. On the morphosyntactic level, the left-hand constituent can be a noun only, which is specified as an item of interest on the semantic level. The item of interest is referred to either directly, as in *mangahead* (manga enthusiast) or *mathhead* (mathematics enthusiast; both AAM/UD), or more commonly through a metonymic reference, as in *petrolhead* (car enthusiast; AAM/OED) or *seamhead* (baseball enthusiast; AAM/Wiktionary).

As stated above, we may abstract a specific subschema for music enthusiasts (9), instantiated by lexemes such as *jazzhead* (AAM/Wiktionary), *hip-hophead*, *metalhead*, and *rockhead* (all AAM/OED).

(9) Semantics: [PERSON WHO IS ENTHUSIASTIC ABOUT (MUSIC STYLE)$_{\mathrm{Y}}$]$_{\mathrm{Z}}$
Morphosyntax: [$_{\mathrm{N}}$N$_{\mathrm{Y}}$ N$_{\mathrm{X}}$]$_{\mathrm{Z}}$
Phonology: /ˈ...$_{\mathrm{Y}}$ hed$_{\mathrm{X}}$/$_{\mathrm{Z}}$

The notation of this subschema differs from the general subschema for enthusiasts only in the specification of the left-hand constituent on the semantic level, namely MUSIC STYLE. Peripheral members of the series are devotees to individual musicians or music bands, such as *Machead* (Mac Miller's fan; AAM/UD) or *Deadhead* (The Grateful Dead band's fan; OED). Also peripherally, the reference to a music style may also be metonymic, as in *rivethead* (heavy metal enthusiast; AAM/UD).

5.2 Natural organisms

The series of bahuvrihi compounds with *head* as the right-hand constituent denoting natural organisms comprises 101 lexemes in the dataset. The distribution among individual types of organisms is given in Table 3.

The schema abstracted from the series is (10).

(10) Semantics: [NATURAL ORGANISM WHOSE (BODY PART)$_{\mathrm{X}}$ HAS FEATURE (Y)$_{\mathrm{Y}}$]$_{\mathrm{Z}}$
Morphosyntax: [$_{\mathrm{N}}$N/A$_{\mathrm{Y}}$ N$_{\mathrm{X}}$]$_{\mathrm{Z}}$
Phonology: /ˈ...$_{\mathrm{Y}}$ ...$_{\mathrm{X}}$/$_{\mathrm{Z}}$

The lexemes in the series are terms used for biological classification, as such being stylistically neutral. As stated above, the lexemes are uniform in their structure and semantics, so there is no reason to assume that each type of organism would form its own series instantiating a specific subschema.

The right-hand constituent is a variable since the series also comprises lexemes whose right-hand constituent is a different body part than *head*, such as *longtail* (bird; OED), *blueleg* (mushroom; Wikipedia), and *silverback* (gorilla; OED). The body parts in these lexemes are literal in animals and metaphorical in plants, referring to their fruits or other upper parts. The left-hand constituents predominantly refer to a static physical feature of the organisms, both literally and metaphorically, such as

Table 3 Frequency of distribution of types of organisms with examples

	Freq.	%	Examples
Fish	36	36	*bighead* (carp), *steelhead* (trout; both AAM/OED)
Birds	25	25	*bufflehead* (duck), *hammerhead* (crane; both AAM/OED)
Mammals	8	8	*doghead* (baboon; OED), *lionhead* (rabbit; AAM/Wikipedia)
Snakes	5	4	*blunthead, copperhead* (both AAM/OED)
Other animals	8	8	*loggerhead* (turtle), *roughhead* (iguana; both AAM/OED)
Plants	19	19	*dragonhead, elephanthead* (both AAM/Wikipedia)
Total	101	100	

size (*bighead*, a fish; AAM/OED), colour (literally *bluehead*, a fish; metaphorically *steelhead*, a fish; both AAM/OED), shape (literally *roundhead*, a fish; metaphorically *hammerhead*, a shark; both AAM/OED), and a lack of plumage (*baldhead*, a bird; AAM/Wiktionary).

5.3 Inanimate referents

The series of bahuvrihi compounds with *head* as the right-hand constituent denoting inanimate referents comprises 63 lexemes in the dataset. Within the series, however, we can also abstract other specific subseries, namely those denoting fasteners, such as screws, bolts, and nails, fish lures, Harley-Davidson engines, and types of acne. The distribution of the subseries is given in Table 4.

The general schema for inanimate referents abstracted from the series is (11).

(11) Semantics: [INANIMATE REFERENT WHOSE (PART)$_X$ HAS FEATURE (Y)$_Y$]$_Z$
Morphosyntax: $[_N N/A_Y\ N_X]_Z$
Phonology: /ˈ...$_Y$ ...$_X$/$_Z$

All the lexemes in the series denoting inanimate referents are stylistically neutral. The right-hand constituent is a variable, as it may be instantiated with other lexemes than *head*, as in *hardtop* (car; OED), *paperback* (book; OED), and *greenback* (banknote; OED). However, a random search for other parts as right-hand constituents denoting inanimate referents shows that these are rare, which may be given by a high salience of upper parts of referents, metaphorically expressed as *head*.

As given in Table 4, lexemes denoting inanimate referents include four specific subseries.

5.3.1 Fasteners

This specific series comprises 30 lexemes, such as *domehead* (Google), *starhead* (Wikipedia), and *roundhead* (AAM/OED), denoting screws, bolts, and nails. The subschema abstracted from the subseries is (12).

(12) Semantics: [FASTENER WHOSE (PART)$_X$ HAS FEATURE (Y)$_Y$]$_Z$
Morphosyntax: $[_N N/A_Y\ N_X]_Z$
Phonology: /ˈ...$_Y$ ...$_X$/$_Z$

Table 4 Frequency of distribution of (sub)schemas for inanimate referents with examples

	Freq.	%	Examples
Inanimate referents – miscellaneous	11	17	*anglehead* (torch), *twinhead* (pump; both AAM/Google)
Fasteners	30	47	*starhead* (Wikipedia), *roundhead* (AAM/OED)
Fish lures	13	21	*beadhead, hardhead* (both AAM/Google)
Harley-Davidson engines	6	10	*knucklehead, shovelhead* (both AAM/Wikipedia)
Types of acne	3	5	*blackhead, whitehead* (both AAM/OED)
Total	63	100	

Terms for fasteners form a series of such substantial size and specificity that we may assume that they instantiate a specific subschema. The right-hand constituent is a variable, as it may also be instantiated with terms denoting other parts, as in *coarse thread*, *sharp-point*, and *ring-shank* (all Google). The left-hand constituent prototypically denotes the shape of the whole upper part, such as *domehead* or *roundhead*, or the pattern in the upper part, such as *starhead* or *hexhead* (AAM/Google). Peripherally, it may also refer to other features, such as the inventor as in *phillipshead* (Wikipedia).

5.3.2 Fish lures

There are 13 lexemes forming the series, such as *beadhead, hardhead* (both AAM/Google), and *wobblehead* (Google), in the dataset. The specific subschema for fish lures abstracted from the series is (13).

(13) Semantics: [FISH LURE WHOSE HEAD$_X$ HAS FEATURE (Y)$_Y$]$_Z$
Morphosyntax: [$_N$N/A/V$_Y$ N$_X$]$_Z$
Phonology: /ˈ...$_Y$ hed$_X$/$_Z$

The specific subschema (13) overlaps with the schema for natural organisms (10) with the right-hand constituent as a variable (cf. *redtail* as another term for a fish lure; Google) since fish lures are meant to imitate water organisms, such as small fish or water insects. However, we can assume an independent status of subschema (13) with the right-hand constituent fixed as *head* since, for instance, *woolhead* (AAM/Google) refers to a lure that is made of wool as a whole and not just its "head" part, and *wobblehead* and *jighead* (AAM/Google) also refer to the movement of the whole lure in water. This might also suggest that the link (notated by index X) between the semantic levels and the other two levels is weakened (cf., "there is probably a cline of relatedness between [...] lexemes used as independent words and their bound use" (Booij, 2010: 58)). The left-hand constituent is motivated by the colour of the "head" part (*pinkhead*; Google) or the material of the "head" part (*beadhead*; AAM/Google) or of the whole lure (*woolhead*). The left-hand constituent may also be conceptualized dynamically on the basis of its movement in water, which leads to the possibility of instantiating the left-hand variable with a verb, as in *jighead, swinghead* (Google), and *wobblehead* (also cf. the deverbal adjective in *shakyhead*; AAM/Google).

5.3.3 Harley-Davidson engines

A rather interesting instance of an emergence of a specific subschema can be found within the sphere of Harley-Davidson engines. Six terms for engines produced by the Harley-Davidson company have the morphological structure following that of subschema (14).

(14) Semantics: [HARLEY-DAVIDSON ENGINE WHOSE HEAD$_X$ HAS FEATURE (Y)$_Y$]$_Z$
Morphosyntax: [$_N$N/A$_Y$ N$_X$]$_Z$
Phonology: /ˈ…$_Y$ hed$_X$/$_Z$

Diachronically, the first terms coined, *F-head* (Wikipedia) and *flathead* (AAM/ Wikipedia), were used for the respective types of engines by various engine manufactures, apparently by making use of the general schema for inanimate referents (11). However, the pattern later caught on for coining names of specifically Harley-Davidson engines, namely *ironhead, knucklehead, panhead*, and *shovelhead* (all AAM/Wikipedia) thus establishing an "internal" schema for this brand of motorcycles.

The fixed right-hand constituent *head* refers to the engine's cylinder head or its cover. The left-hand constituents refer either literally, *flathead*, or metaphorically, *knucklehead*, to the shape of the head, with the exception of *ironhead* referring to the material used.

5.3.4 Types of acne

The series from which subschema (15) is abstracted comprises three lexemes only, namely *blackhead, whitehead* (both AAM/OED), and *yellowhead* (AAM/Google).

(15) Semantics: [TYPE OF ACNE WHOSE HEAD$_X$ HAS (COLOUR)$_Y$]$_Z$
Morphosyntax: [$_N$A$_Y$ N$_X$]$_Z$
Phonology: /ˈ…$_Y$ hed$_X$/$_Z$

Given the highly specific category of the referents, differing from all the other members of the inanimate category in not being man-made objects, and given the fact that they are well-established lexemes (the first two listed in OED), thus exhibiting a relative prominence, we may consider it to form a separate schema. The right-hand constituent is fixed and refers to the top of acne. Out of the subschemas for inanimate referents, subschema (15) is the most restrictive as the left-hand variable is instantiated with adjectives denoting the colour of the top part only.

6 Relational function of schemas

For Construction Morphology, a "cousin" theory of Relational Morphology (see Jackendoff & Audring, 2020b), Booij states that one of its central tenets is "that the lexicon of natural languages is highly structured by means of schemas and subschemas. Morphological schemas, which specify systematic relationships between the form

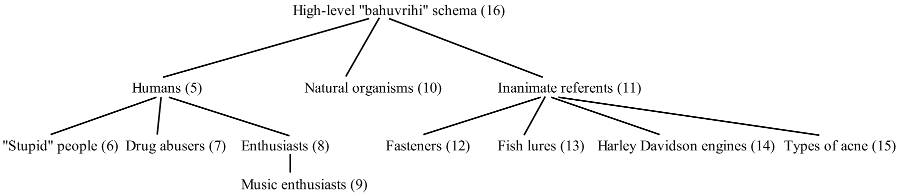

Fig. 1 Taxonomy of schemas abstracted from the dataset

and meaning of sets of complex words, dominate the complex words that are formed according to these schemas" (Booij, 2012: 344).

This section aims to summarize what implications the dataset and the abstracted schemas have for the structure of this specific "island" in the lexicon.

Firstly, the schemas prove to be categories comprising fully specified lexemes. In accordance with the Prototype Theory (Rosch, 1978), the members of a schema demonstrate different degrees of centrality; in other words, schemas contain prototypical as well as peripheral members. For example, the general schema for humans (5) prototypically comprises lexemes denoting human referents and having derogatory connotations. Peripherally, however, the category also includes lexemes denoting non-human referents that are, nevertheless, connected to the category through some anthropomorphic features (toys, cartoon characters) and that lack the derogatory character. A different type of peripherality is demonstrated in schema (7) denoting drug abusers in that the variable in the left-hand position is prototypically instantiated with a lexeme denoting the substance abused but the series also includes a small number of peripheral items not following the pattern. Prototypical members, adhering to all properties of the schema, thus constitute instantiations of the schema, and peripheral members, suppressing some of the schema's properties, are extensions from the prototype (see Langacker, 2013: 17–18).

Secondly, also in accordance with the Prototype Theory, the schemas prove to have fuzzy boundaries. The fuzzy boundaries exhibit themselves as overlaps between two schemas, where the lexemes may belong to two same-level schemas simultaneously. An overlap of two schemas is exhibited, for instance, between schemas (7) and (8), denoting drug abusers and enthusiasts, respectively, as, e.g., *coffeehead* may belong to both categories as someone addicted to coffee or as someone enthusiastic about coffee. A fuzzy boundary is also apparent between schemas of different branches in their taxonomy (see Fig. 1), as exemplified by an overlap between natural organisms (10) and fish lures (13).

Thirdly, the schemas abstracted from the dataset are hierarchically structured. Figure 1 shows the suggested taxonomy.

The abstraction over the (sub)series of lexemes in the dataset has led to 11 schemas. The most comprehensive level in the hierarchy of the schemas is that of (5), (10), and (11). The schemas at this level share the characteristics that the right-hand constituents are variables, referring directly or metonymically to a (body) part, and the notation on the semantic level is underspecified. The justification for distinguishing the schemas at this level is the category of referents they denote, namely humans, natural organisms, and inanimate referents, the mostly derogatory nature of (5), and the classificatory function in natural taxonomy in (10).

Among the varied lexemes instantiating the schemas at this level, schemas (5) and (11) also exhibit specific subseries, leading to subschemas (6)–(8) and (12)–(15), respectively. Since, as mentioned in Sect. 2, the emergence of a schema may be "bottom-up", relatively narrow schemas giving rise to broader ones, and "top-down", constructing a narrower schema inside a broader one (see Jackendoff & Audring, 2020a: 227), we can ascribe the emergence of (6)–(8) and (12)–(15) to the "top-down" formation in that critical numbers of lexemes sharing the same semantics have led to more specific subschemas. The same perspective can be ascribed to the emergence of (9), as a specification of more general (8).

Moreover, all the lexemes in the dataset are also seen as instances of "bahuvrihi" compounds, a type of exocentric compounds with the right-hand constituent denoting a part of the whole. The fact that they all are seen as one type of compounds leads to the assumption that they, as a group, are also represented in the lexicon by a high-level abstract schema, (16).

(16) Semantics: [THING WHOSE (PART)$_X$ IS ASSOCIATED WITH (Y)$_Y$]$_Z$
Morphosyntax: [$_N$N/A$_Y$ N$_X$]$_Z$
Phonology: /ˈ. . .$_Y$ . . .$_X$/$_Z$

The category THING on the semantic level is to be understood as a basic cognitive category corresponding to the word-class of nouns. The emergence of (16) is "bottom-up", being an abstraction over other more specific schemas. Schema (16) is instrumental in the relational function in that it distinguishes this type of compounds from other exocentric, e.g. *scarecrow*, and endocentric compounds. The level of schemas (5), (10), and (11) can thus be seen as the basic level, from which other schemas emerge, both in the "top-down" and "bottom-up" manner.

Fourthly, it is also the conceptualization patterns of individual lexemes that may be instrumental in providing structure to the lexicon. Since, according to the full-entry theory (see Jackendoff & Audring, 2020a: 65), the fully specified lexemes continue to be stored in the memory after the emergence of a schema, their established patterns of conceptualization, which are not otherwise reflected on the semantic level of the abstracted schemas, group together specific lexemes. This is apparent in the specific subseries denoting members of ethnic groups (see Sect. 5.1), which does not have its own specific schema but where the unifying aspect is the shared MEMBER OF CATEGORY FOR CATEGORY metonymy. Also consider, for instance, the microseries *breezehead, windhead*, and *galehead*, which may be too small and peripheral to form its own schema (apparently formed through analogy), but which are related through the same manner of conceptualization.

Lastly, another feature that contributes to the "texture" of the lexicon is a schema's resting activation. Schemas' resting activation is "correlated with the frequency with which they are used" (Jackendoff & Audring, 2016: 484) and as such expresses the relative strength of the schema in the lexicon. However, the analysis of the dataset cannot provide any, not even approximate, estimate of the schemas' resting activation. As "the frequency of a schema is the sum of the frequencies of its instances" (Jackendoff & Audring, 2016: 484), the estimate of the resting activation of the schema would have to be based on the estimated frequencies of *all* lexemes instantiating the schema by identifying the number of their tokens in the corpus. The dataset cannot provide such estimates because the series whose corresponding schemas have variables in the right-hand positions are incomplete (do not include lexemes with other parts than *head*) and because the corpus only served as a source of forms, which were subsequently paired with all recorded meanings in dictionaries to include meanings not attested in the corpus (see Sect. 3), with the consequence that the dataset includes pairings of form and meaning beyond those in the corpus.

7 Generative function of schemas

On the distinction between the relational and the generative roles of schemas, Jackendoff and Audring proposed the Relational Hypothesis, worded as "*all* schemas can be used relationally, but only some – the productive schemas – can be used generatively" (Jackendoff & Audring, 2020a: 54, emphasis in original). Thus, an existence of a schema does not automatically mean that the schema is productive. Whether or not a schema is productive, i.e., whether it can be used generatively, is not a feature of the schema as a whole but rather of its variables as "productivity amounts to the openness of the variable, where we take openness to mean the degree to which it accepts new instantiations" (Jackendoff & Audring, 2020a: 41).

The dataset analysed for the present study suggests that productivity of schemas is not of a binary but rather of a gradient nature. An example of a schema that appears to be non-productive is that for types of acne (15). What limits the openness of the variable in the left-hand position, however, is not its semantic specification but the non-existence of other referents which this highly specific subschema could denote. The openness of the variable is thus given extra-linguistically. From the same perspective, we may consider schemas for natural organisms (10) and fasteners (12), among others, as semi-productive since the openness of the respective variables to accept new instantiations depends on a discovery of a new organism or on the invention of a new type of fastener, respectively. As, especially in the case of natural organisms, a discovery of new organisms is rather rare, the productivity of the respective schemas is highly limited. Schemas that appear to be fully productive are those for human referents (5)–(9). Here the openness of the variables seems to have little limitation extra-linguistically as, for instance, any human feature can serve for derogatory reference in the case of schema (5), or people can become enthusiastic about a variety of things in the case of schema (8). The high productivity of schemas for human referents is also reflected in the high occurrence of their respective lexemes in crowdsourced dictionaries, such as UD, which especially lists nonce-formations or otherwise peripheral items.

Also, if we assume that coiners opt for the most specific schema available for the formation of a lexeme, i.e., the one that best specifies the semantic properties of the concept to be named (the bottom-up approach), high-level schema (16) is not likely to be instrumental in coining new lexemes (the generative function) and has the relational function only.

As stated in Sect. 2, in the Relational Morphology conception, schemas have a declarative character, which means that they are stored as templates in the memory. In the generation of new lexemes, the variables in the schemas are instantiated with existing lexemes retrieved from the lexicon through the process of unification.

However, the Relational Morphology theory does not deal with the question of what principles determine which lexeme is to be retrieved from the lexicon in order to achieve the required lexical meaning of the resulting word. Since the lexemes in the dataset denote specific concepts, the function of the generative role of the corresponding schemas is that of naming, i.e., providing names for existing concepts. As, for instance, *flathead* may denote, among other things, a fish, a "stupid" person, a motorcycle engine, and a screw, each lexeme (of this identical form) instantiating a different (sub)schema, we may hardly presuppose the existence of semantically underspecified *flathead* whose lexical meaning would later be specified in its use; in other words, that we would first coin the word *flathead* and then search for a referent with a flat head. The lexical meaning of a word is thus not a secondary idiosyncratic shift of the structural one (see Sect. 2 for the distinction between the two types of meaning), but the concept, i.e., the substrate for the future lexical meaning, stands at the beginning of the naming process.

Consequently, naming presupposes initial mental processing of the concept, namely categorization and selection of a feature characterising the concept. The concept is represented in the mind of the speaker as Idealized Cognitive Model (ICM; see Lakoff, 1987: 147), which is a mental structure that organizes our knowledge and understanding of the concept. During the process of conceptualization, the concept "is associated with a complex ICM (source), which provides the basis for naming the thing (target). Second, guided by language-independent factors such as salience, economy, and metonymy, only certain components of the complex ICM get selected and named by a given speech community" (Radden & Panther, 2004: 8). The result of conceptualization is then paired with an existing suitable schema whose variables are instantiated with lexemes representing the result of the conceptualization (see Kos, 2023, 2024). It is the structural meaning of a schema that determines its suitability for the result of conceptualization, as its category and semantic specification (including the semantic restrictions of the variables) must match. So, in naming the shark *Sphyrna tiburo*, for example, its resulting conceptualization, which can be paraphrased as "it is a type of shark (category) and, among other things, the shape of its head resembles that of a bonnet (specification)", can thus be matched with schema (10) since the schema's category on the semantic level is the superordinate NATURAL ORGANISM, its variable X is specified as BODY PART, and its variable Y is underspecified on the semantic level but restricted to nouns or adjectives on the morphosyntactic level. The variables in the schema are then instantiated with lexemes denoting the respective parts of conceptualization, which leads to the emergence of the lexeme *bonnethead* (OED). Since the category specified on the semantic level has no interface link to the other levels of the schema, it remains unexpressed.

Nevertheless, the lexicon may contain more schemas suitable for the same conceptualization, which may lead to synonymous terms, as another attested term for this shark, *bonnet shark* (Wikipedia), suggests. So, the same conceptualization can also be matched with schema (4) for endocentric NN compounds, where the category (indexed as Y) and the variable X are semantically underspecified but restricted to nouns on the morphosyntactic level, their relation also being underspecified. The variable Y is then instantiated with the lexeme *shark* representing the category, and the variable X is instantiated with the lexeme *bonnet* denoting the most salient part of the specification (cf. the low salience of HEAD since all sharks have heads), which leads to the emergence of the lexeme *bonnet shark*.

However, conceptualization and schema selection are not necessarily strictly sequential, the latter necessarily following the former, since a pre-selected schema may affect, or direct, the way speakers mentally process the named concept. For instance, "bahuvrihi" schemas, such as (5), appear to be the only ones in English that refer to humans and are straightforwardly derogatory, so if we want to coin a term for a human with derogatory connotation, (5) is at hand, with the consequence that the schema dictates that the right-hand constituent be a body part. In, e.g., *bighead* (conceited person, someone who has a very high opinion of how important they are; AAM/OED), the salient parts of the ICM that are used for naming are IMPORTANT, OPINION, and ANNOYING. The feature ANNOYING prompts the selection of (5), as a schema for forming derogatory terms for humans, so the feature OPINION needs to undergo further conceptualization through the metonymic chain ACTIVITY FOR RESULT (more specifically THINKING FOR OPINION) and PART OF BODY FOR ACTIVITY (more specifically HEAD FOR THINKING); in other words, the initial feature OPINION needs to appear in the resulting conceptualization as HEAD in order to match the pre-selected schema. The feature IMPORTANT is further conceptualized as BIG through the conceptual metaphor IMPORTANCE IS LARGE SIZE.

Also, as mentioned in Sect. 2, semantic features of individual lexemes that are too idiosyncratic to be abstracted in the corresponding schema are retained in the memory together with the fully specified lexemes. The possible consequence for the generation of new lexemes is that a manner of conceptualization of existing lexemes, which otherwise is not part of the semantic level of a schema, can have an impact on the way we mentally approach the newly named concept. In our dataset, this has become most apparent in derogatory terms for "stupid" people (see Sect. 5.1.1), where the lack of intelligence can be seen as slowness, hardness, emptiness, etc., and these established patterns of conceptualization may serve as templates of the formation of new lexemes of that kind.

In general, a new lexeme is formed on a model, a lexical item, stored in the lexicon. The model lexical item can be not only a schema but also a single word; in other words, apart from making use of schemas, we can also coin new lexemes by analogy based on association with specific lexemes. The formation based on schemas and on analogy are complementary strategies (see Booij, 2010: 89), and analogy on one specific lexeme cannot be ruled out even if there is an existing productive schema available in the lexicon since "analogical formations are not necessarily devoid of rules" (Mattiello, 2017: 54). In terms of the present study, apart from apparent instances of analogy in the formation of *tighthead* (first attestation in 1959, see OED) on the model of *loosehead* (first attestation in 1907, see OED; both denoting types of rugby players) and *breezehead, windhead*, and *galehead* (which lexeme served as a model for the others cannot be identified; all taken from UD, denoting people of different speeds of thinking), whether individual lexemes in larger series instantiating an established schema were formed on the model of the schema or on analogy on one of the lexemes cannot be determined.

Moreover, analogy may also trigger emergence of a new schema as "analogical word formation may develop [. . .] into a pattern that abstracts away from specific model words" (Booij, 2010: 90).

8 Conclusion

The present study investigates a specific sample of data, namely English bahuvrihi compounds with the right-hand constituent *head*, with the aim of identifying the variety of schemas that can be abstracted from the sample and to discuss implications relevant for the structure of that part of lexicon and for their role in the generation of new lexemes. The findings may even contribute to understanding the nature of schemas beyond the studied dataset.

The dataset comprises 330 forms (type frequency), and the pairing with their attested meanings in dictionaries gives rise to 540 lexemes (type frequency) for analysis. The abstraction over the specific series in the dataset has led to 11 different tentative schemas. Together with the high-level schema abstracted from these specific schemas, the number of schemas arising from the dataset totals 12.

The parameters applied in this study for assuming that a schema may be abstracted from a series of lexemes is the possibility to schematize a unified meaning on the semantic level (since otherwise all lexemes in the dataset share the same form), the size of the series, and, in the case of series of small size, the prominence of lexemes comprising the series, which is given both by the distinctness of the category they belong to from categories of lexemes of a higher-level series and by the assumed frequency of use, i.e., their resting activation.

The notation of the semantic level of both fully specified lexemes and schemas in the present approach includes the category to which the lexical item belongs. Even if it has no representation on the morphosyntactic and phonological levels, thus being formally unexpressed in fully specified lexemes, the category contributes to structuring the lexicon through relational links grouping together lexemes that share the same category within the relational function of schemas and is instrumental in matching the initial conceptualization of the named concept with a suitable schema within their generative function.

The right-hand positions of the schemas may be semantically restricted variables limiting the openness of the variable to a part of the whole, fixed and literal constituents, which are fully specified but have not lost their semantic value, and fixed and idiomatic constituents, i.e., affixoids, which have no direct literal representation on the semantic level. The left-hand positions of schemas can be either semantically open or semantically restricted variables.

The schemas are hierarchically structured, and their taxonomy suggests that we can identify 4 levels of specificity. The topmost level, the tentative beginner in the hierarchy, is the most general schema for bahuvrihi compounds, abstracted in the "bottom-up" manner from the level below. Being the most abstract one, this schema has the relational function only, not serving as a template for the generation of novel lexemes of this type. The level below the topmost one can be seen as the basic level in the taxonomy since it is the level from which other schemas are abstracted, either the most abstract one in the "bottom-up" manner or the lower-level ones in the "top-down" manner.

Another aspect emerging from the dataset is that schemas prove to be categories comprising both prototypical and peripheral members. At the same time, they have fuzzy boundaries as individual lexemes may instantiate two series simultaneously, the consequence of which is an overlap of the two schemas.

The study also suggests that apart from the features notated in the schemas, it is also the semantics of the fully specified lexemes that contributes to structuring the lexicon by grouping together lexemes on the basis of their conceptualization. The specific manners of the conceptualization of lexemes instantiating individual schemas may also be instrumental in the generation of novel lexemes by providing conceptual patterns for the mental processing of concepts to be named.

As for their generative role, the schemas abstracted from the dataset prove to have a naming function, i.e., they are instrumental in naming existing concepts. It is thus a concept, the conceptual substrate for the future lexical meaning of the newly coined lexeme, that stands at the beginning of the naming process. The concept is first categorized and conceptualized by selecting some aspect(s) of the whole concept for the naming purposes. The result of conceptualization is then paired with a matching schema retrieved from the lexicon. As more than one schema may match the result of conceptualization, a competition of schemas may occur. It is also argued that a pre-selected specific schema may have an impact on the conceptualization process.

Finally, the productivity of the schemas in the present study proves to be limited merely extra-linguistically as the only factor that restricts the openness of the variables to accept new instantiations appears to be the existence of (new) concepts to be named within a given category.

Funding information Open access publishing supported by the institutions participating in the CzechELib Transformative Agreement. The author did not receive support from any organization for the submitted work.

The author has no relevant financial or non-financial interests to disclose.

Declarations

Competing interests The author declares no competing interests.

References

Barcelona, A. (2008). The interaction of metonymy and metaphor in the meaning and form of "bahuvrihi" compounds. *Annual Review of Cognitive Linguistics*, *6*(1), 208–281. https://doi.org/10.1075/arcl.6.10bar.

Bauer, L., & Huddleston, R. (2002). Lexical word-formation. In R. Huddleston & G. K. Pullum (Eds.), *The Cambridge grammar of the English language* (pp. 1621–1722). Cambridge: Cambridge University Press.

Benko, V. (2014). Aranea: Yet another family of (comparable) web corpora. In P. Sojka, A. Horák, I. Kopeček, & K. Pala (Eds.), *Text, speech and dialogue (TSD 2014, LNAI 8655)* (pp. 257–264). Berlin: Springer. https://doi.org/10.1007/978-3-319-10816-2_31.

Booij, G. (2010). *Construction morphology*. London: Oxford University Press.
Booij, G. (2012). Construction morphology, a brief introduction. In *Morphology* (Vol. 22, pp. 343–346). Berlin: Springer. https://doi.org/10.1007/s11525-012-9209-x.
Jackendoff, R. (2002). *Foundations of language*. London: Oxford University Press.
Jackendoff, R., & Audring, J. (2016). Morphological schemas: Theoretical and psycholinguistic issues. *The Mental Lexicon*, *11*(3), 467–493. https://doi.org/10.1075/ml.11.3.06jac.
Jackendoff, R., & Audring, J. (2020a). *The texture of the lexicon: Relational morphology and the parallel architecture*. London: Oxford University Press.
Jackendoff, R., & Audring, J. (2020b). Relational Morphology: A Cousin of Construction Grammar. *Frontiers in Psychology*, 11. https://doi.org/10.3389/fpsyg.2020.02241.
Kos, P. (2023). The role of metonymy in naming: If longhair then apple tree and teacher. *Review of Cognitive Linguistics*, *21*(1), 86–114. https://doi.org/10.1075/rcl.00128.kos.
Kos, P. (2024). Instantiating variables in schemas within Relational Morphology. *Linguistica Pragensia*, *34*(2), 77–99. https://doi.org/10.14712/18059635.2024.2.1.
Lakoff, G. (1987). *Women, fire, and dangerous things: What categories reveal about the mind*. Chicago: University of Chicago Press.
Langacker, R. W. (1999). Correspondences, compositionality, and grammar. *Manusya: Journal of Humanities*, *2*(2), 61–76.
Langacker, R. W. (2013). *Essentials of cognitive grammar*. Oxford: Oxford University Press.
Langacker, R. W. (1987, 1991). *Foundations of Cognitive Grammar*. Stanford: Stanford University Press.
Mattiello, E. (2017). *Analogy in word-formation*. Berlin: de Gruyter.
Radden, G., & Panther, K. U. (2004). Introduction: Reflections on motivation. In G. Radden & K. U. Panther (Eds.), *Studies in linguistic motivation* (pp. 1–46). Berlin: de Gruyter.
Rosch, E. (1978). Principles of categorization. In E. Rosch & B. Lloyd (Eds.), *Cognition and categorization* (pp. 27–48). Hillsdale: Lawrence Erlbaum Associates.